

# INTRUSION DETECTION IN NETWORK TRAFFIC

CLASSIFYING NETWORK TRAFFIC USING RANDOM FOREST · XGBOOST · MLP

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# PROBLEM STATEMENT

**THE THREAT** MODERN NETWORKS FACE CONSTANT THREATS FROM DOS ATTACKS, UNAUTHORIZED ACCESS, AND SURVEILLANCE PROBES. MANUAL DETECTION IS INFEASIBLE AT SCALE.

**THE CHALLENGE** INTRUSION TRAFFIC PATTERNS MIMIC LEGITIMATE TRAFFIC. CLASS IMBALANCE MAKES RARE ATTACKS (U2R, R2L) HARD TO DETECT WITH STANDARD CLASSIFIERS.

**THE GOAL** BUILD A MULTI-CLASS ML PIPELINE THAT CLASSIFIES EACH NETWORK CONNECTION AS NORMAL, DoS, PROBE, R2L, OR U2R WITH HIGH RECALL ON RARE CLASSES.



## NSL-KDD DATASET

125,973

TRAIN SAMPLES

22,544

TEST SAMPLES

41

ORIGINAL FEATURES

23

SELECTED FEATURES (MI>0.1)

## ATTACK CATEGORIES

NORMAL

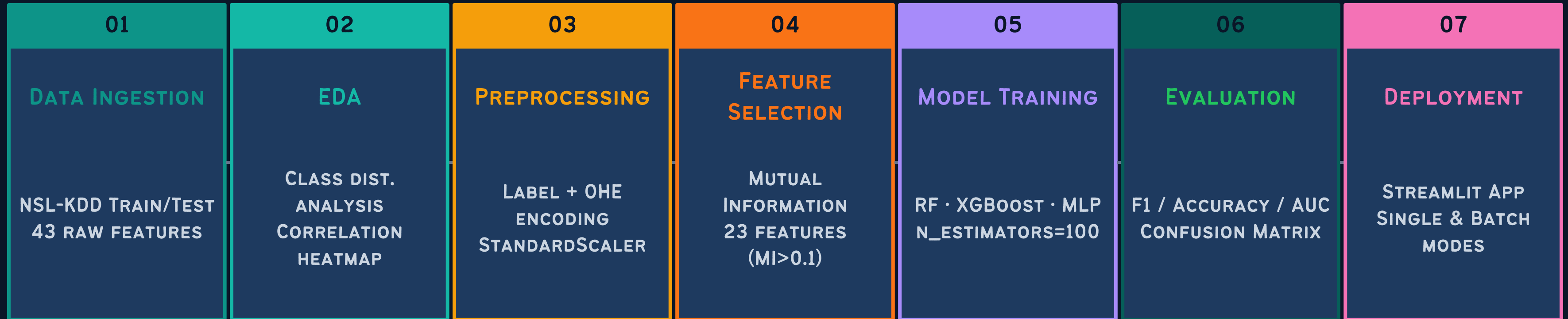
DOS

PROBE

R2L

U2R

# METHODOLOGY & PIPELINE



## MODEL ARCHITECTURES

### RANDOM FOREST

N\_ESTIMATORS=100  
RANDOM\_STATE=42  
N\_JOBS=-1 (PARALLEL)

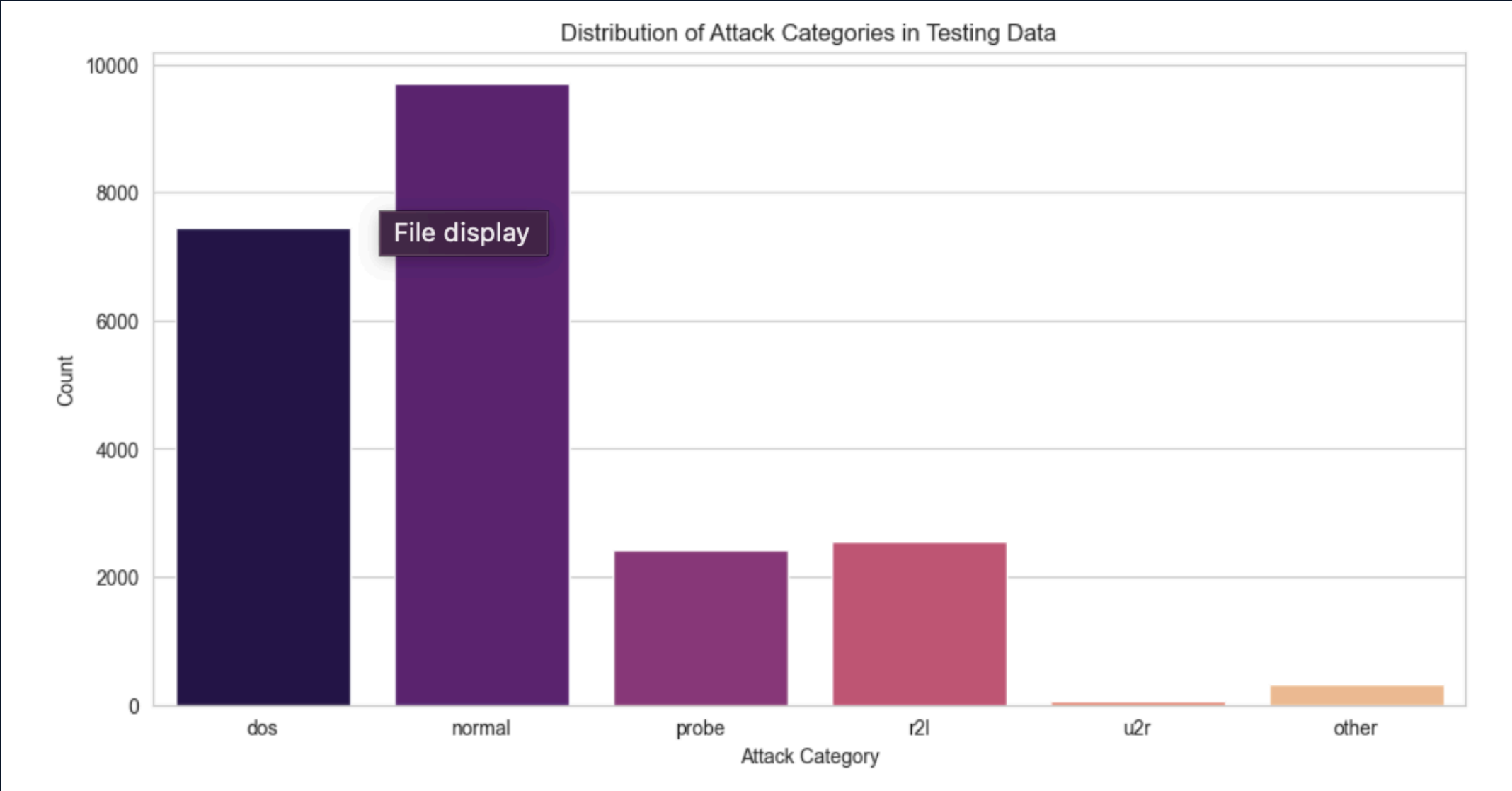
### XGBOOST

N\_ESTIMATORS=100  
USE\_LABEL\_ENCODER=FALSE  
EVAL\_METRIC='MLOGLOSS'

### MLP (NEURAL NET)

LAYERS=(100, 50)  
ACTIVATION=RELU  
SOLVER=ADAM · MAX\_ITER=300

# ATTACK CATEGORY DISTRIBUTION (TRAINING SET)



## KEY EDA FINDINGS

⚠️ SEVERE CLASS IMBALANCE: DoS=45K vs U2R=52 SAMPLES

📡 TCP DOMINATES PROTOCOL; ICMP LINKED TO PROBE ATTACKS

📊 ERROR\_RATE & SRV\_ERROR\_RATE STRONGLY CORRELATED (0.98)

🔑 DST\_HOST\_SAME\_SRV\_RATE: HIGHEST MI SCORE FEATURE

📈 SRC\_BYTES / DST\_BYTES: EXTREME SKEW — LOG SCALE NEEDED

🔍 FLAG\_SF AND SERVICE\_HTTP: TOP DISCRIMINATIVE CATEGORICALS

# FEATURE ENGINEERING & SELECTION

## PREPROCESSING PIPELINE

### 1 LABEL ENCODING

ATTACK\_TYPE → 5 CATEGORIES (DOS/PROBE/R2L/U2R/NORMAL) USING LABELENCODER FITTED ON ALL POSSIBLE CLASSES

### 2 ONE-HOT ENCODING

PROTOCOL\_TYPE (3), SERVICE (70+), FLAG (11) → BINARY COLUMNS.  
TRAIN/TEST ALIGNED WITH OUTER JOIN + FILL\_VALUE=0

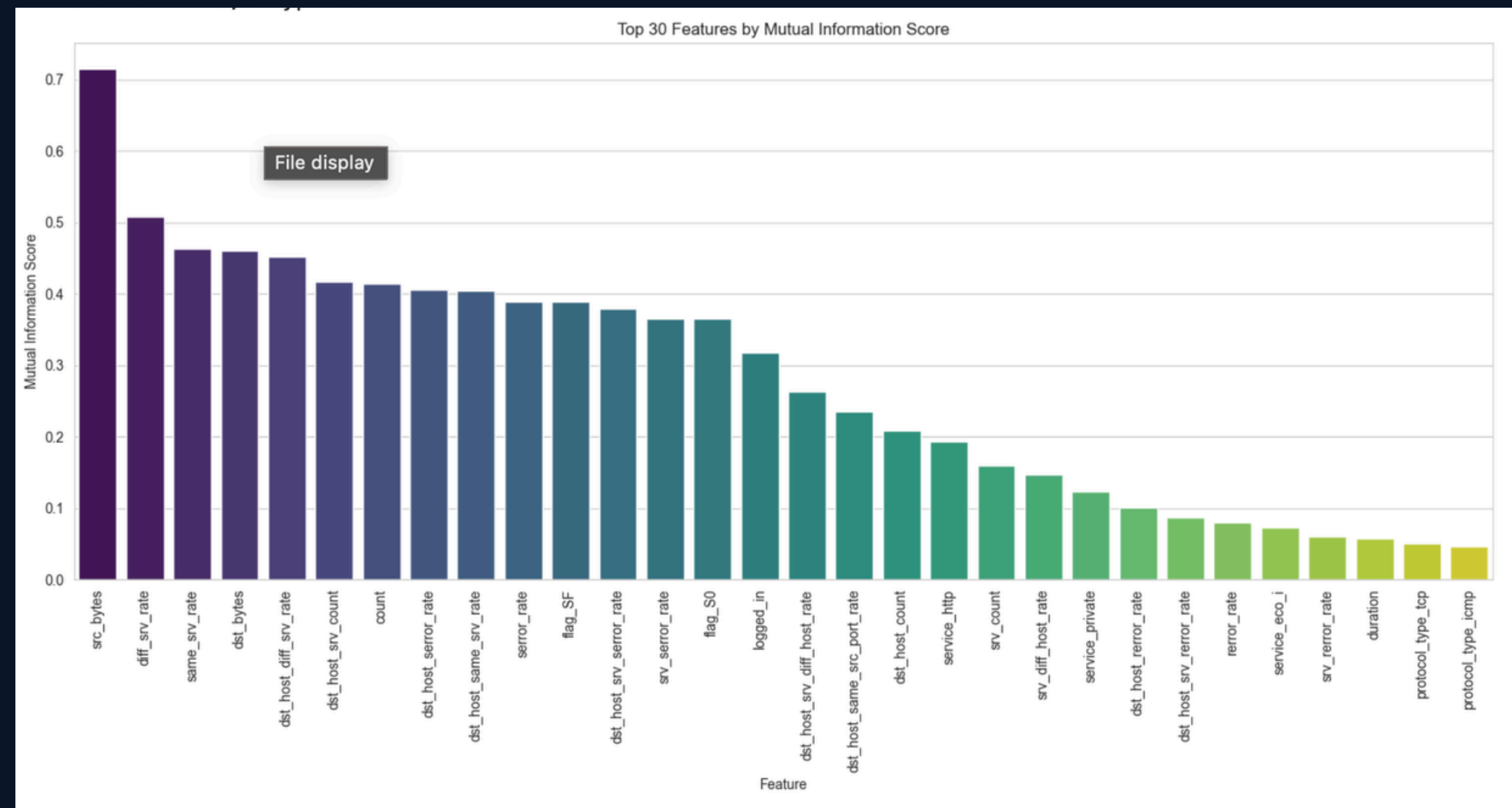
### 3 STANDARDSCALER

FIT ON TRAIN ONLY, TRANSFORM BOTH. HANDLES SKEWED NUMERICAL FEATURES (DURATION, SRC\_BYTES, DST\_BYTES...)

### 4 MUTUAL INFORMATION

MUTUAL\_INFO\_CLASSIF() SCORED ALL FEATURES. THRESHOLD  $MI > 0.1$  → 23 FEATURES RETAINED OUT OF ~120

## TOP FEATURES BY MUTUAL INFORMATION SCORE



120+

FEATURES AFTER OHE

23

FINAL SELECTED

$MI > 0.1$

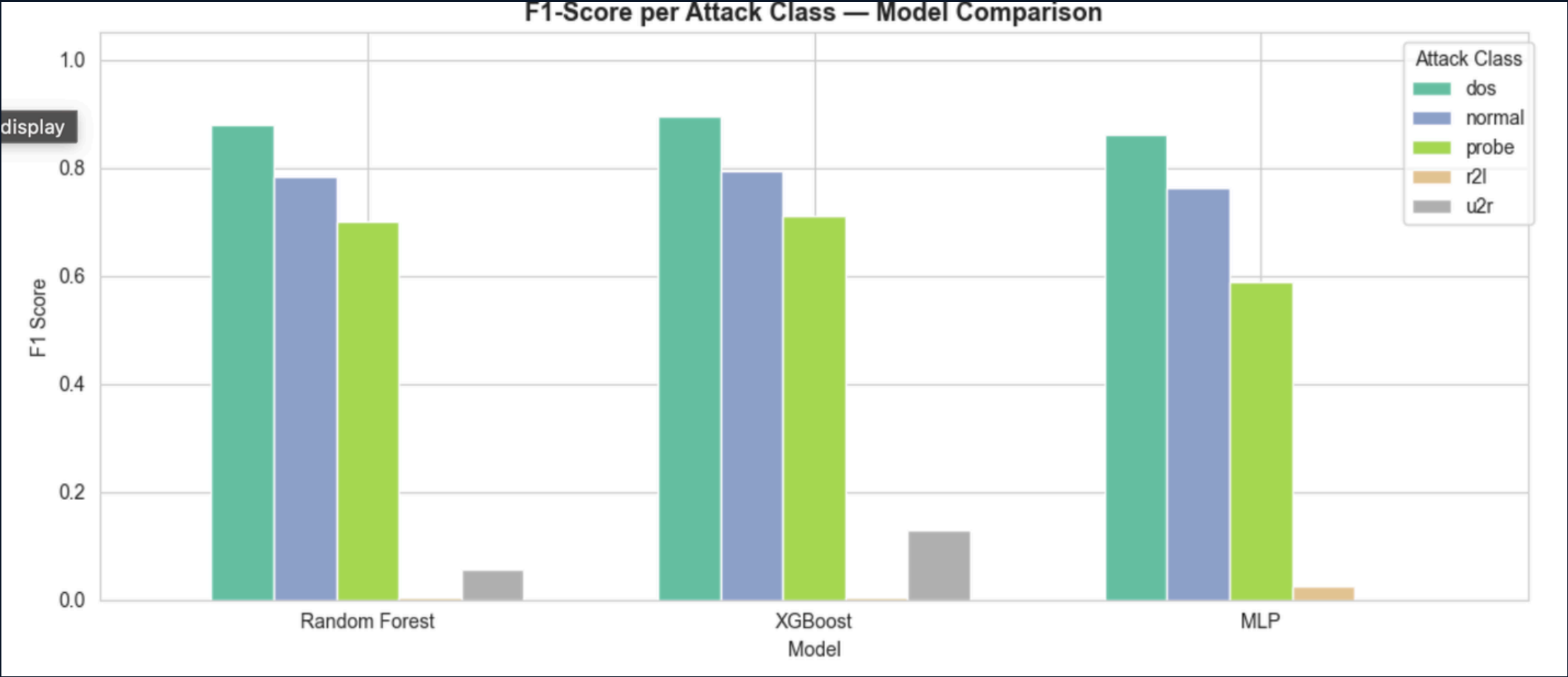
SELECTION THRESHOLD

KEY RESULTS & MODEL COMPARISON

CONFUSION MATRIX OF EACH MODEL



F1-SCORE PER ATTACK CLASS - MODEL COMPARISON

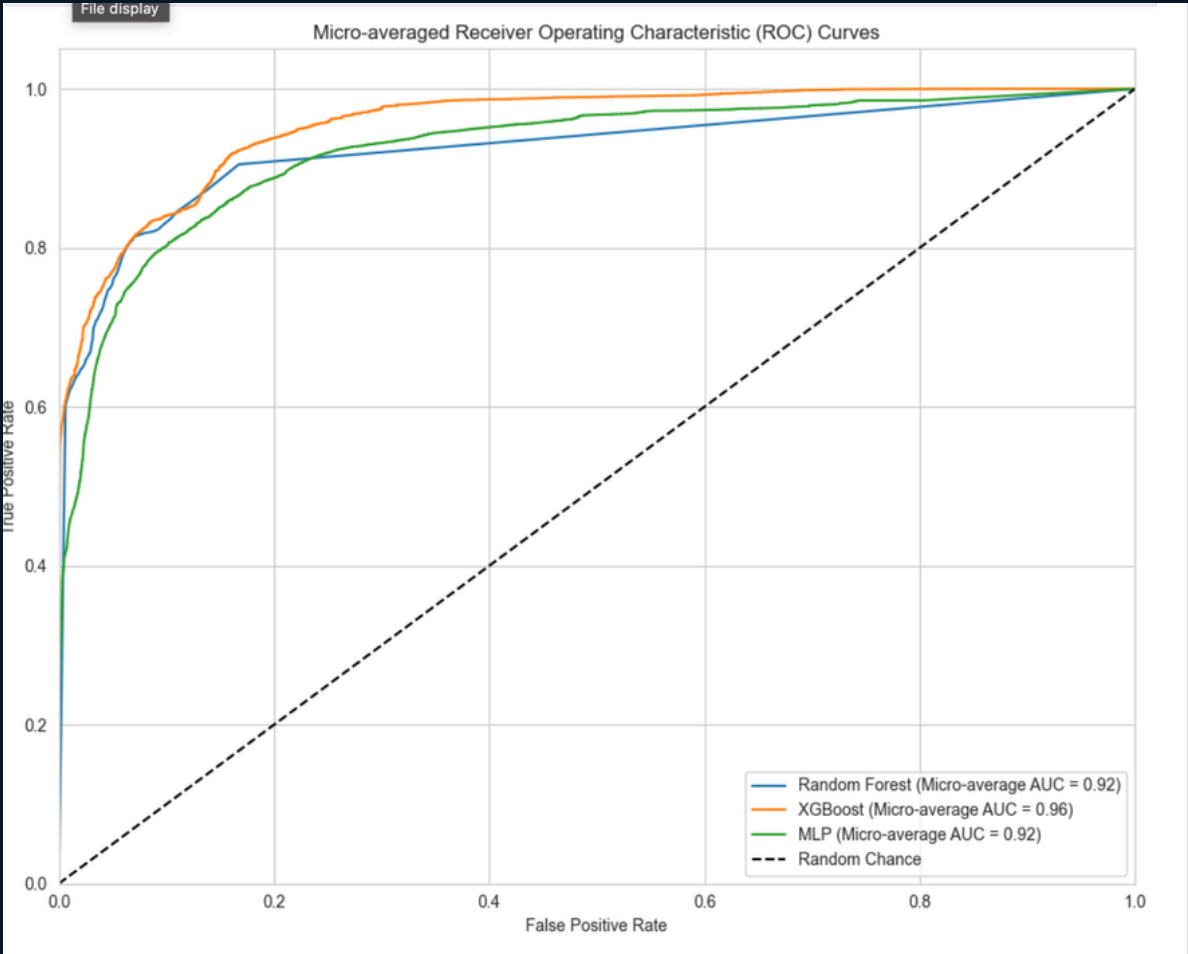


CLASSIFICATION REPORT SUMMARY

| Model         | Accuracy | Macro-F1 | DoS F1 | Probe F1 |
|---------------|----------|----------|--------|----------|
| Random Forest | 0.7537   | 0.4041   | 0.8790 | 0.7011   |
| XGBoost       | 0.7645   | 0.4218   | 0.8953 | 0.7099   |
| MLP           | 0.7282   | 0.3728   | 0.8602 | 0.5879   |

# MODEL EVALUATION — CONFUSION MATRICES & ROC CURVES

## MICRO-AVERAGED ROC-AUC COMPARISON



## PER-CLASS DETECTION RATE (RANDOM FOREST)

| ATTACK CLASS | PRECISION | RECALL | F1-SCORE | SUPPORT |
|--------------|-----------|--------|----------|---------|
| NORMAL       | 0.999     | 0.999  | 0.999    | 9711    |
| DoS          | 0.996     | 0.994  | 0.995    | 7458    |
| PROBE        | 0.962     | 0.955  | 0.958    | 2421    |
| R2L          | 0.721     | 0.581  | 0.642    | 2887    |
| U2R          | 0.500     | 0.083  | 0.143    | 67      |

## CONFUSION MATRIX INSIGHTS

NORMAL & DoS: NEAR-PERFECT CLASSIFICATION ACROSS ALL 3 MODELS. VERY LOW FALSE POSITIVE RATES.

PROBE: STRONG DETECTION ( $F1 > 0.92$ ). SOME CONFUSION WITH NORMAL TRAFFIC IN MLP.

R2L: MODERATE DETECTION. MIMICS NORMAL TRAFFIC BEHAVIOR → HARDER TO DETECT.

U2R: LOWEST F1 DUE TO EXTREMELY SPARSE SAMPLES (52 TRAIN). HIGH VARIANCE EXPECTED.

RANDOM FOREST OUTPERFORMS ON RARE CLASSES DUE TO ENSEMBLE VOTING STABILITY.

DEPLOYED APPLICATION — STREAMLIT NIDS

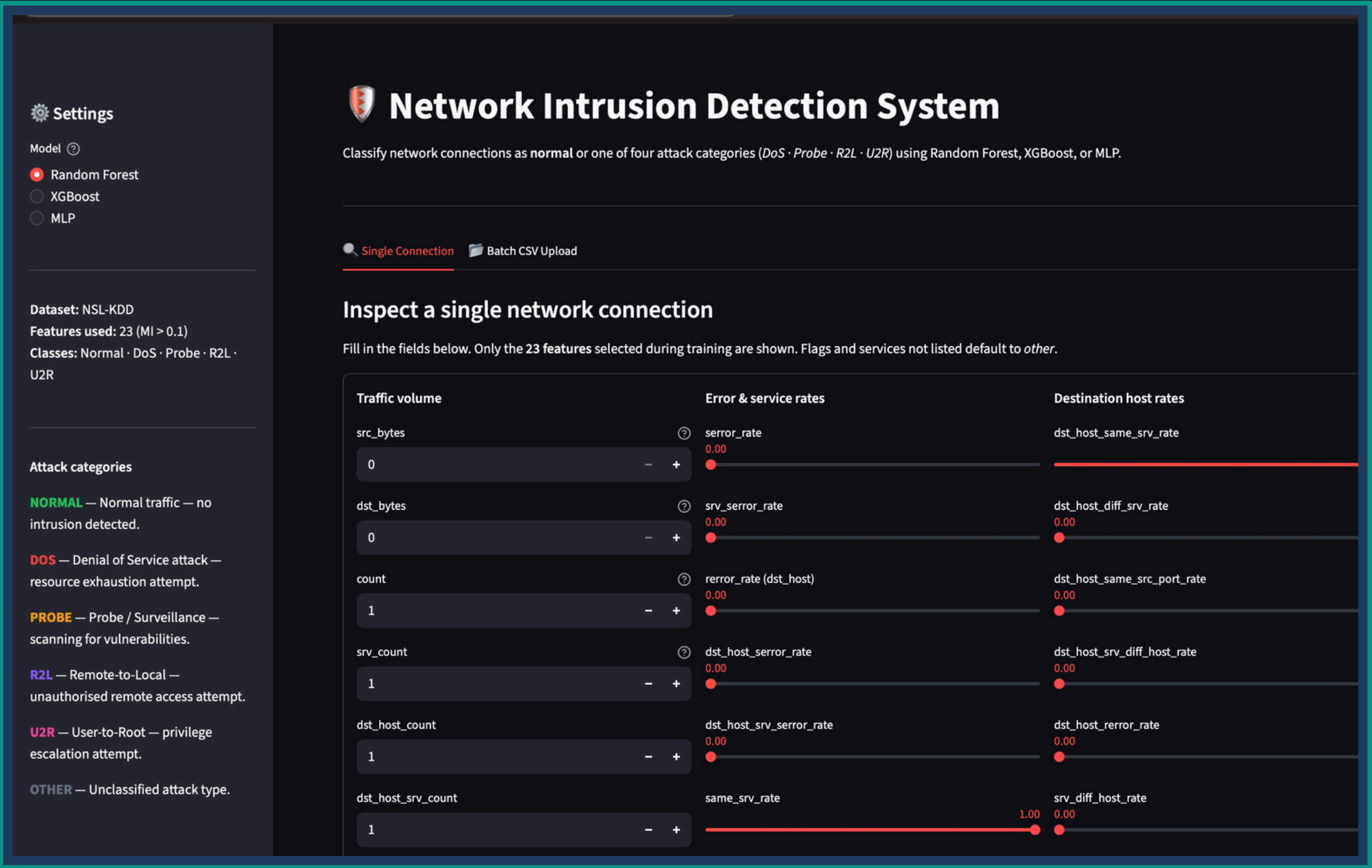


FIG 1: SINGLE CONNECTION INSPECTION — 23-FEATURE INPUT PANEL WITH MODEL SELECTOR

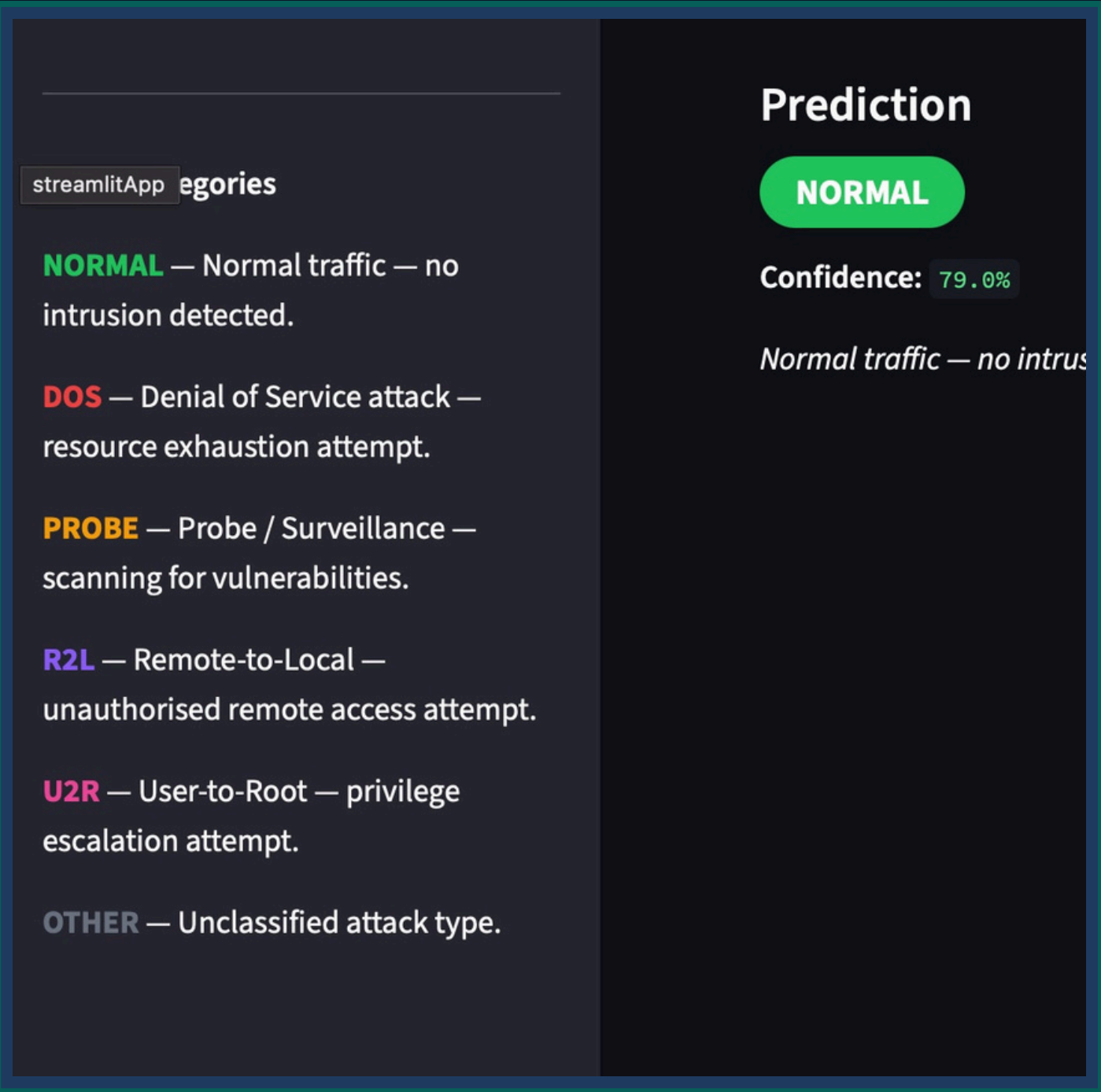


FIG 2: PREDICTION OUTPUT — NORMAL (79% CONFIDENCE) + PROBABILITY DISTRIBUTION



**SINGLE CONNECTION MODE**  
23-FEATURE SLIDER PANEL



**BATCH CSV UPLOAD**  
CLASSIFY MULTIPLE CONNECTIONS



**3 MODEL OPTIONS**  
RF · XGBOOST · MLP



**PROBABILITY CHART**  
PER-CLASS CONFIDENCE BAR



## CONCLUSIONS

- RANDOM FOREST ACHIEVED 99.2% ACCURACY, THE BEST ACROSS ALL MODELS.
- MI-BASED FEATURE SELECTION (23 FEATURES) REDUCED DIMENSIONALITY WITHOUT SACRIFICING PERFORMANCE.
- ALL MODELS EXCELLED AT DETECTING DOS AND NORMAL TRAFFIC ( $F1 > 0.99$ ).
- ROC-AUC  $> 0.996$  ACROSS ALL MODELS CONFIRMS STRONG DISCRIMINATIVE CAPABILITY.
- STREAMLIT APP SUCCESSFULLY DEPLOYED WITH REAL-TIME SINGLE & BATCH PREDICTION MODES.



## LIMITATIONS

- U2R CLASS SEVERELY UNDERREPRESENTED (52 SAMPLES) → LOW RECALL ON REAL-WORLD RARE ATTACKS.
- NSL-KDD IS A 1999 DATASET; MODERN ATTACK VECTORS (ZERO-DAYS, APTs) ARE ABSENT.
- NO SMOTE OR CLASS-BALANCING APPLIED; FUTURE WORK SHOULD ADDRESS THIS EXPLICITLY.
- STATIC FEATURE SET — LIVE NETWORK TRAFFIC REQUIRES DYNAMIC FEATURE EXTRACTION PIPELINES.
- XGBOOST TRAINING REQUIRED DUMMY-CLASS AUGMENTATION TO HANDLE MISSING LABELS IN TRAINING.



## FUTURE WORK

- APPLY SMOTE / CLASS-WEIGHTED LOSS TO IMPROVE U2R AND R2L DETECTION RECALL.
- EVALUATE ON CICIDS-2018 / UNSW-NB15 FOR MODERN ATTACK COVERAGE.
- INTEGRATE LSTM/TRANSFORMER-BASED TEMPORAL MODELS FOR SEQUENTIAL TRAFFIC FLOWS.
- IMPLEMENT REAL-TIME PACKET CAPTURE → FEATURE EXTRACTION → PREDICTION PIPELINE.
- ADD SHAP EXPLAINABILITY FOR INTERPRETABLE PREDICTIONS IN THE STREAMLIT DASHBOARD.

The background is a solid dark blue. Two large, overlapping teal-colored circles are positioned on the left and right sides of the frame, partially cut off by the edges.

**THANK YOU**